

CHRISTMAS DECORATION LIGHT



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Here we present a circuit for LED-based Christmas decoration light. This decoration light is not only heartwarming, it is brilliant and cool too. The circuit can be applied in many ways, and there is plenty of scope for you to show your creativity in producing a design that is truly unique.

The basic circuit uses a set of five LEDs that light up according to mounting on a festival decoration of your choice. You can buy a readymade decoration piece, or design and make your own. It can be hung on the Christmas tree or elsewhere in the room.

PARTS LIST

Semiconductors:

- IC1 - 7555 timer
- IC2 - CD4040 binary counter
- IC3-IC8 - 74HC164 shift register
- IC9 - UM66 music generator
- IC10 - 7805, 5V regulator
- T1, T2 - 2N3904 npn transistor
- LED1-LED61 - 5mm LED
- D1-D4 - 1N5408

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon):

- R1, R2 - 1-kilo-ohm
- R3, R66 - 470-ohm
- R4-R63 - 220-ohm
- R64 - 1.5-kilo-ohm
- R65 - 330-ohm
- R67 - 390-ohm
- VR1 - 20-kilo-ohm preset

Capacitors:

- C1 - 100 μ F, 16V electrolytic
- C3, C4 - 10 μ F, 16V electrolytic
- C2 - 0.01 μ F ceramic disk
- C5 - 1000 μ F, 25V electrolytic
- C6 - 0.1 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 9V, 1.5A secondary transformer
- LS1 - 8-ohm, 0.5W speaker

Fig. 1: Circuit for Christmas decoration light

